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Vaccine Adjuvants

Facile Preparation of Carbohydrate-Containing Adjuvants Based on Self-Assembling Glycopeptide Conjugates

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Abstract: Carbohydrates are intriguing biomolecules possessing diverse biological activities, including immune stimulating capability. However, their biomedical applications have been limited by their complex and heterogeneous structures. In this study, we have utilized a self-assembling glycopeptide conjugate (GPC) system to produce uniform nanoribbons appending homogeneous oligosaccharides with multivalency. This system successfully translates the nontrivial structural differences of oligomannoses into varied binding affinities to C-type lectin receptors (CLRs). We have shown that GPCs could promote the CLR-mediated endocytosis of ovalbumin (OVA) antigen, and two mannotriose-modified peptides **F3m2** and **F3m5** exhibit potent activity in inducing antigen-presenting cell maturation, as indicated by increased CD86 and MHCII expression. In vivo studies demonstrated that GPCs, combined with OVA antigen, significantly enhanced OVA-specific antibody production. Specifically, **F3m2** and **F3m5** exhibited the highest immunostimulatory effects, eliciting both Th1- and Th2-biased immune responses and promoting differentiation of $CD4^+$ and $CD8^+$ T cells. These findings highlight the potential of GPCs as vaccine adjuvants, and showcase their versatility in exploiting the biological functions of carbohydrates.

Introduction

Vaccines are a potent therapeutic intervention that modifies the immune system to prevent or treat infectious diseases and cancer.[1] While peptide and oligosaccharide antigens have shown promise in vaccine development, their poor immunogenicity compared to inactivated viruses limits their clinical application.[2] To address this issue, various adjuvants have been developed, including metal adjuvants,[3] engineering proteins,[4] oil emulsion,[5] synthetic polymers,[6] microorganisms and their metabolites.[7] These adjuvants can activate the innate immune response or assist antigen uptake by antigen-presenting cells,[8] thereby improving vaccine efficacy. However, only a few adjuvants have been approved by regulatory agencies for use in humans due to concerns about their safety, mechanism of action, structural complexity, and self-immunogenicity.[9] Therefore, the search for new adjuvant candidates represents a promising avenue for discovering safe and effective vaccine adjuvants.

Carbohydrates are ubiquitous biomolecules that play a pivotal role in various immune system events.[10] The oligomannose-containing high mannose-type *N*-glycans, which are expressed on yeast and fungi, can be recognized by a variety of host innate immune receptors, including C-type lectin receptors (CLR), such as DC-SIGN, Dectin-2, MR, and Langerin.[11] The multivalent binding of oligomannose with receptors mediates the endocytosis of pathogens and the production of immune factors by immune cells, followed by the activation of the immune system.[11c,12] Although these immunostimulating abilities make carbohydrates attractive candidates for vaccine adjuvants, the structural complexity and heterogeneity of carbohydrates from natural sources pose extreme challenges to understanding their action mechanisms and applying them as therapeutics.[13]

To fully exploit the advantages of carbohydrates, extensive research has been conducted to develop their mimics that are more structurally maneuverable. These systems often require covalently constructed frameworks for loading the oligosaccharides to establish the clusters,[14] where polymers[14c] or dendrimers[14b] are necessary to be introduced. In contrast, using self-assembling strategies to display glycan structures may convert simpler and more structurally editable monomers into supramolecules as bioactive carbohydrate mimics. For instance, Stupp et al. utilized sulfated lipoglycopeptides to form supramolecular nanostructures that mimic natural glycosaminoglycans (GAGs).[15] Such a system can significantly enhance the signal transduction of bone morphogenetic proteins and promote bone regeneration in the spine. Chen et al. constructed dendritic peptide scaffolds to hitchhike three oligosaccharides and assessed their ability to induce cytokine release from immune cells.[16] Our group also discovered that monosaccharide-modified self-assembling peptides could form glycocluster-like nanoparticles.[17] Despite these advances, examples using simple self-assembling structures to mimic polysaccharides remain limited, facing the challenge that the change of glycan structures can dramatically alter the morphology of

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formed nanostructures,[18] which may influence their activity.[19]

In this study, we systematically evaluated a diphenylalanine-derived[20] glycopeptide system that can be modified with a variety of oligosaccharides, including *O*-linked oligomannoses, *N*-linked *N*-acetylglucosamine (GlcNAc), *N*,*N*′-diacetyl chitobiose (GlcNAcβ1-4GlcNAc), and complex-type glycans. Our findings reveal that an optimized peptide structure could effectively carry various oligomannoses, and self-assemble to form stable nanoribbons, which showed varying affinities to CLR proteins in an *in vitro* binding assay. Additionally, several mannosylated peptides demonstrated robust immune stimulatory effects *in vivo* (Figure 1). The mechanism of action and biocompatibility of these glycopeptide conjugates (GPCs) as vaccine adjuvant candidates have also been investigated.

Results and Discussion

Syntheses of GPCs

In light of the vital immunological roles of high mannose-type *N*-glycans, we wonder whether a robust system could be developed to amplify the distinctions of oligomannose epitopes within the glycan structure through multivalent presentation and interrogate the potentially varied biological impacts.[21] Drawing inspiration from our previous study,[17] we envisioned that self-assembling glycopeptide conjugates may serve as a foundation for constructing nanostructures that mimic the multivalent interactions of glycans. Accordingly, we initiated the study by examining a previously optimized sequence containing a tri-phenylalanine core, a mannosylated *C*-termini, and a hydrophilic phosphoryl serine that had been proved essential for successful self-assembly[17] (Figure 2a). Employing the Fmoc-based solid phase peptide synthesis (SPPS),[22] we synthesized glycopeptide sequences bearing α-mannose (**A1m**), α-1,2-mannobiose (**A2m**), and α-1,2-α-1,2-mannotriose (**A3m**). While the

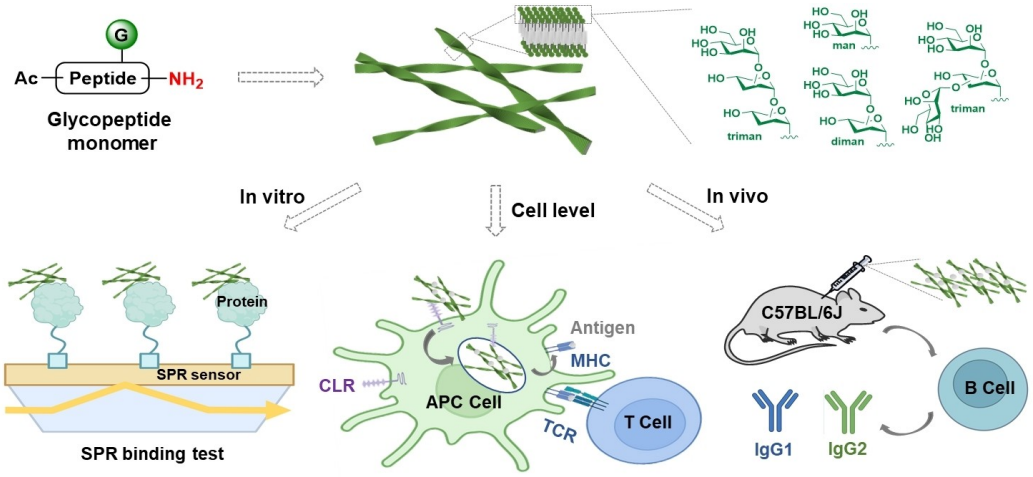


Figure 1. Application of self-assembling glycopeptide conjugates in exploring the structure–activity relationships of oligosaccharides and evaluating potential immunostimulating ability.

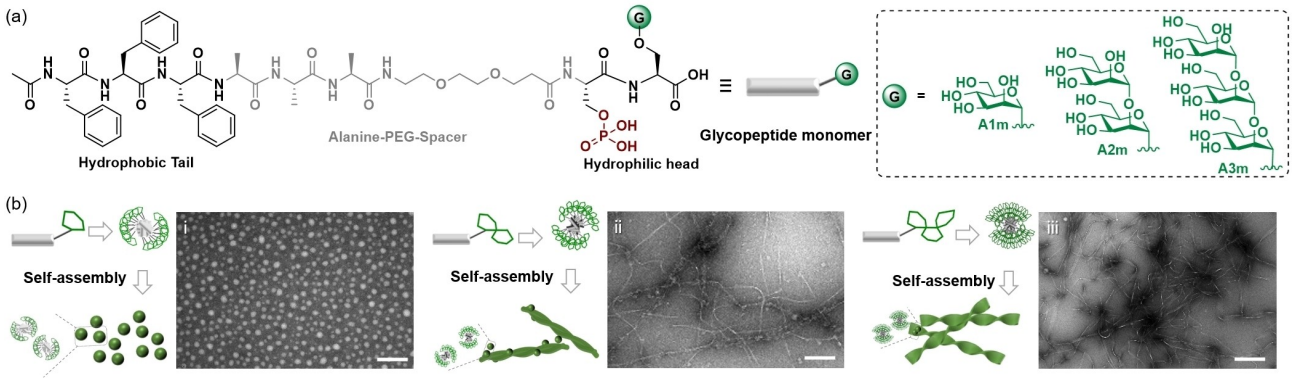


Figure 2. Structures and self-assembling properties of model GPCs. a) Sequence of peptide conjugates **A1m**, **A2m**, and **A3m** synthesized by SPPS. b) TEM images and illustration of the nanostructure formed in water by incubating 3 mM of i) **A1m**, ii) **A2m**, or iii) **A3m**. Scale bar: 200 nm.

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mannosylated peptide **A1m** self-assembled into nanospheres, as reported previously (Figure 2b), we observed that neither the dimannose- nor the trimannose-modified peptides formed uniform spherical structure. Instead, nanofibrils or nanoribbons were displayed under transmission electron microscopy (TEM) following 72 h incubation of the glycopeptides in water (Figure 2b; see also Figure S1a in the Supporting Information). In addition, Thioflavin T (ThT) fluorescence assays indicated gradual aggregation of the mannosylated peptides **A2m** and **A3m**, suggesting distinct morphological changes compared to **A1m** (Figure S2). Given that the morphology of nanostructures can impact their properties, including biological activities,[19] a more robust glycopeptide sequence capable of producing consistent nanostructures, preferably stable nanoribbons, upon functionalization with various oligosaccharides is required for the evaluation of a broader range of glycans.

In pursuit of this objective, we modified the sequence of the mannosylated peptide **A1m** (Table 1, entry 1) with aim of achieving the desired nanoribbon formation by incorporating amino acid residues containing alkyl side chains, such as norleucine (Nle).[15,23] However, replacing the phenylalanine (Phe) residues with Nle (**B1m** in entry 2, and **C1m** in entry 3), or increasing the Nle content in the sequences (**D1m** in entry 4 and **E1m** in entry 5), did not influence the resulting GPCs to produce stable nanostructures. Considering that the self-assembling ability of amphiphilic peptides can be impacted by both their hydrophobic and hydrophilic termini, we turned to adjust the hydrophilicity of the C-terminus by changing the carboxylate to an amide (**F1m**, entry 6). To our delight, the modified sequence **F1m** successfully generated nanoribbon structures after incubation in water for 24 h, as evidenced by the TEM images (see below, Figure 4b). These results suggest that the additional

Table 1: GPC sequences screened and evaluated for their self-assembling ability. The morphology of samples refers to visualization by the naked eye or TEM after incubating the corresponding GPC in water. *O*-Phosphorylation is represented by a red ball and α-*O*-mannose by a green ball for clarity.

Entry	Sequences	Morphology
1	A1m: Ac–Phe-Phe-Phe-Ala-Ala-Ala-PEG-Ser-Ser–OH	Nanoparticle
2	B1m: Ac–Nle-Phe-Phe-Ala-Ala-Ala-PEG-Ser-Ser–OH	Irregular accumulation
3	C1m: Ac–Nle-Nle-Phe-Ala-Ala-Ala-PEG-Ser-Ser–OH	Irregular accumulation
4	D1m: Ac–Phe-Phe-Nle-Nle-Ala-Ala-PEG-Ser-Ser–OH	Poor solubility
5	E1m: Ac–Nle-Nle-Phe-Phe-Ala-Ala-PEG-Ser-Ser–OH	Poor solubility
6	F1m: Ac–Phe-Phe-Phe-Ala-Ala-Ala-PEG-Ser-Ser–NH_2	Twisted nanoribbon

π–π stacking interactions provided by the tri-Phe core in our system are crucial for maintaining peptide self-assembly, in contrast to the hydrophobic Nle. Additionally, it appears that reducing the negative charges in the sequence promotes nanoribbon formation, presumably due to the increased influence of consecutive alanine residues on inducing β-sheet twisting, leading to the production of fibrils and derived nanostructures.[24] At this stage, we have successfully obtained a sequence (**F1m**) that rapidly forms stably nanoribbons and can serve as a template for incorporating various oligosaccharides for further investigations.

Next, we synthesized a range of GPCs to investigate the impact of varied glycosidic linkages in oligosaccharides on their biological activities when presented in a multivalent format. Representative truncated structures from the high mannose-type *N*-glycans were selected as oligosaccharide epitopes (Figure 3), including α-1,2-mannobiose (**F2m1**), α-1,3-mannobiose (**F2m2**), α-1,6-mannobiose (**F2m3**), α-1,2-α-1,2-mannotriose (**F3m1**), α-1,2-α-1,3-mannotriose (**F3m2**), α-1,2-α-1,6-mannotriose (**F3m3**), α-1,3-(α-1,6)-mannotriose (**F3m4**), and α-1,6-α-1,6-mannotriose (**F3m5**). In addition to these six peptide conjugates encompassing *O*-linked oligomannoses, we synthesized four peptides bearing *N*-linked GlcNAc (**F1g**), GlcNAcβ1-4GlcNAc (**F2g**), complex-type asialo- and sialo-oligosaccharide (**F9x** and **F11x**) that represent the epitopes from complex type *N*-glycans, as well as one non-glycosylated peptide (**F0g**). Specifically, the "cassette" strategy was employed to *O*-glycosylated Ser or *N*-glycosylated Asn building blocks directly into the peptide sequences during SPPS, allowing for the preparation of GPCs **F1g**, **F2g**, **F2m1-3** and **F3m1-5** with relatively small-sized glycans (See the Supporting Information for experimental details, Page S30–62). Notably, despite the strong hydrophobicity and steric hindrance of the triphenylalanine core and fully protected glyco-amino acid building blocks in these sequences, the synthetic efficiency remained high. Additionally, FITC labeled glycopeptides (**F0g-FITC**, **F2m1-FITC**, **F3m1-FITC**, **F3m2-FITC**, **F3m4-FITC**, and **F3m5-FITC**) were similarly prepared for visualization of GPCs at the cellular and animal levels (Figures S47–S53). Alternatively, for the complex-type asialo- and sialo-oligosaccharide-modified peptide conjugates **F9x** and **F11x**, we used a chemoenzymatic strategy, converting the GlcNAcylated peptide **F1g** into **F11x** under the Endo-M-catalyzed transglycosylation conditions.[25] Subsequent removal of terminal sialic acids from **F11x** with sialidase yielded GPC **F9x**. These results highlight the convenience, robustness, and modularity of glycopeptide-based amphiphiles compared to other structures such as glycolipids and lipopeptides.

Nanostructure characterization

With the GPCs in hand, we explored their self-assembling properties and characteristics of the supramolecular structures they formed. TEM imaging revealed the formation of nanoscaled twisted-ribbons in mannosylated GPCs **F1m**, **F2m1-3**, and **F3m1-5** after incubation for 24 h in water at a concentration of 3 mM (Figure 4b). CD spectra analysis

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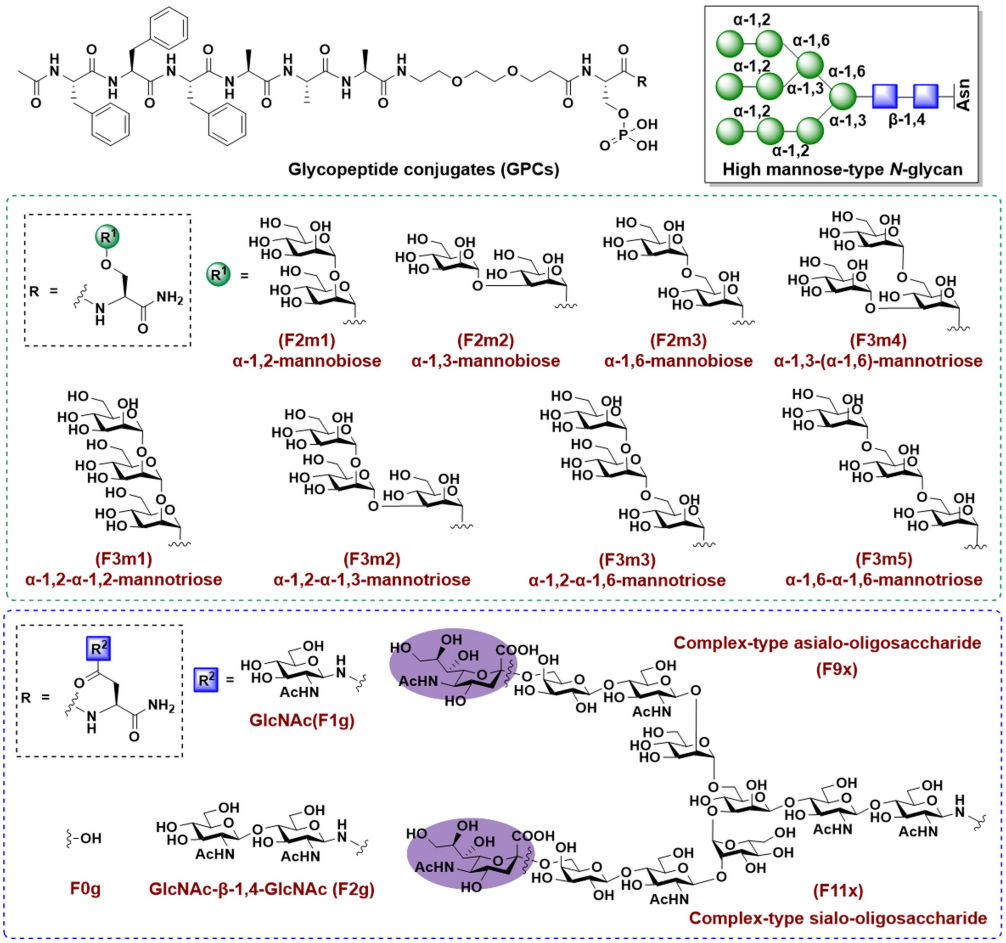


Figure 3. Design of self-assembling glycopeptide conjugates containing varied *O*- or *N*-glycans. A cartoon illustration of high mannose-type *N*-glycan shown in the inset.

demonstrated the presence of β-sheet structures within the self-assemblies, indicated by the positive bands near λ = 195 nm and negative bands near λ = 215 nm (Figure 4c).[24,26] These results indicated the structural similarity among the optimized *O*-mannosylated sequences. The stability of the nanostructures derived from **F1m**, **F2m1**-**3**, and **F3m1**-**5** was evaluated by measuring their zeta potential in H_2O, revealing relatively high and negative zeta potential values (−54.7 to −35.1 mV), indicating good stability (Figure 4d).[27] The negative zeta potential values are likely due to the introduced phosphorylation for sequence stabilization,[15] suggesting that the saccharide moieties are presented on the surface of formed nanostructures (Figure 4a). CD experiments further confirmed that the secondary structures of the self-assembling GPCs, such as **F2m1**, remained largely unchanged under varying pH and temperature conditions, suggesting their potential for biological applications. Furthermore, we explored the ability of this self-assembly system to accommodate other saccharides, including *O*-linked fucose (**F1f**, Figure S43), *N*-linked GlcNAc (**F1g**) and derivatives (**F2g**, **F9x**, and **F11x**). Interestingly, it appears that the *O*-glycosylated sequences tended to form nanoscale twisted-ribbons (Figure S3), while *N*-glycosylation modifications resulted in the formation of nanofibers (Figure 4e,f) or loose nanospheres (Figure 4g,h), suggesting that structural differences at the glycosylation sites may influence the formed supramolecular structures. By comparing the *N*-glycosylated GPC samples, it appeared that an increase in the size of the glycans might reduce the stability of self-assembly, resulting in less compact nanostructures. Nonetheless, the uniform nanostructures generated from *O*-glycosylated self-assembling peptides, such as *O*-mannosylated ones, are suitable for investigating the glycosylation-activity relationships in the subsequent stage of our study.

Protein-binding ability of GPC nanoribbons

The obtained *O*-glycan-modified nanoribbons enabled us to interrogate the precise structural features impacting their protein binding abilities in a multivalent presentation, mimicking the interaction mode of natural glycocalyx.[28] Employing surface plasmon resonance (SPR) spectroscopy, we assessed the binding capacity of mannosylated nanoribbons to five biologically significant proteins, including DC-SIGN,[29] Dectin-2,[30] FIMH,[31] Langerin,[11b] and mannose receptor (MR),[32] known to bind high mannose-type *N*-glycans.[11c] Analysis of the SPR curves revealed a clear

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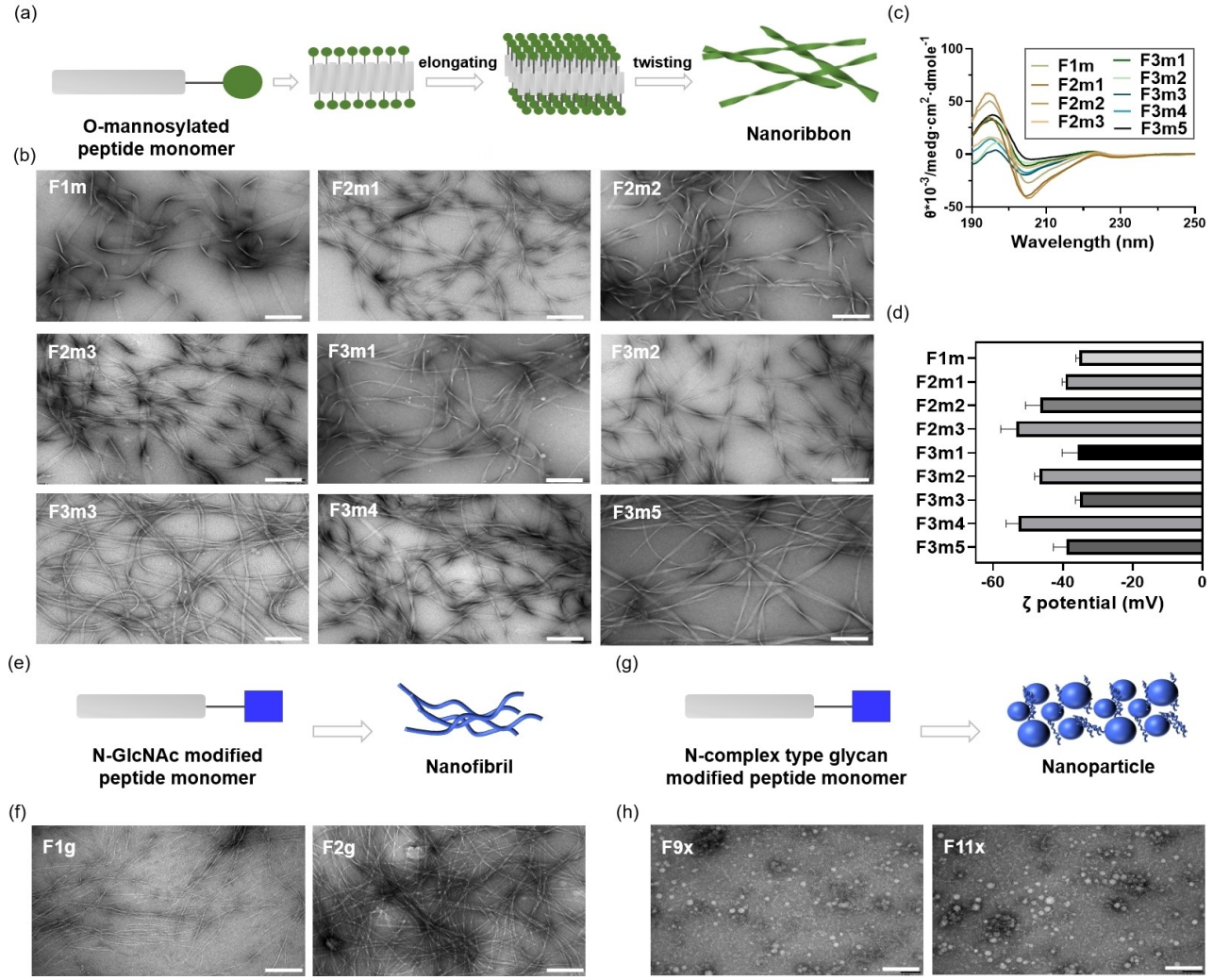


Figure 4. Characterization of nanostructures derived from GPCs. a) Possible mechanisms of nanoribbon formation from GPCs. b) TEM images of twisted-ribbon-like nanostructures formed in 3 mM solutions of GPCs **F1m**, **F2m1-3**, and **F3m1-5**. c) CD spectra and d) zeta potential measurements of the GPCs **F1m**, **F2m1-3**, and **F3m1-5** (150 μM in H_2O, $n=3$). Values are the mean ± sd. e) Illustration and f) TEM images of nanofibrils formed in 3 mM solutions of GPCs **F1g** and **F2g**. g) Illustration and h) TEM images of nanospheres formed in 3 mM solutions of GPCs **F9x** and **F11x**. Scale bar: 200 nm.

concentration-dependent and saturation binding behavior for glycopeptide nanoribbons with the five proteins. In contrast, the non-glycosylated nanoribbons derived from **F0g** exhibited negligible binding, effectively ruling out non-specific interactions as a contributing factor (Figure 5b,c, and Table S1 in the Supporting Information). The equilibrium dissociation constant (K_D) of the glycopeptide-derived nanostructures ranged from 0.39 to 50 μM, demonstrating affinities up to approximately three orders of magnitude higher than unmodified mannose (Table S1). These findings underscore the crucial role of multivalent interactions facilitated by the supramolecular assembly of these glycopeptides.

Previously, Gao et al. employed glycan microarrays to access the binding of various oligomannose structures with glycan-binding proteins.[11c] Their study has shown that DC-SIGN displayed a preference for binding linear α-1,2-α-1,3-mannotriose, whereas FimH exhibited a stronger affinity for mannobiose over mannotriose. Furthermore, their findings indicated that Dectin-2 exhibited a heightened binding affinity toward branched oligomannose structures containing the α-1,3-(α-1,6)-trimannose core. Upon comparing their results with our own data, we observed similar binding preferences for DC-SIGN and FimH (Figures 5a, S6a,e). However, notable differences arose when considering the requisite glycan chain length for detectable protein binding in microarray-based experiments. For instance, our experimental results indicate that DC-SIGN not only favors binding to α-1,2-α-1,3-mannotriose and α-1,6-α-1,6-mannotriose epitopes, but also to the branched α-1,3-(α-1,6)-mannotriose structure. In contrast, microarray-based glycan analysis mainly provides similar information for more extended glycan chains containing these key trimannose motifs. This limitation could potentially complicate data interpretation and, in some instances, lead to deviated conclusions. Additionally, in the case of Dectin-2 (Figures 5a

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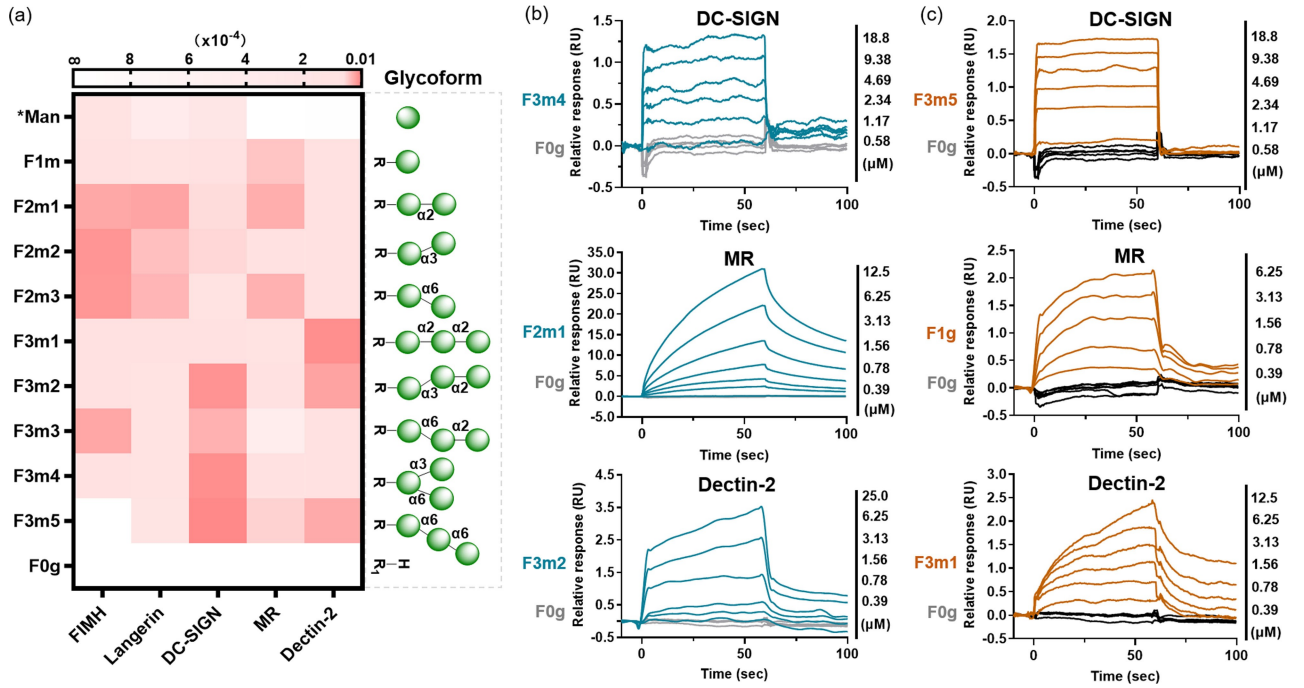


Figure 5. Binding profiles of the GPC-derived nanoribbons towards representative high mannose-binding proteins. a) Heat map presentation of binding affinities between mannose (*Man) or mannosylated nanostructures and representative high-mannose-binding proteins. Mannose and the glycoforms included in the corresponding GPCs are displayed as cartoons on the right. b,c) SPR curves of F0g (gray) and representative GPC-derived nanoribbons (blue and orange) at the indicated concentration binding to immobilized proteins. The GPCs bearing known receptor-binding oligosaccharides are colored in blue and new receptor-binding oligosaccharides discovered in this study are colored in orange. RU, response units.

and S6b), our GPC-based nanostructures exhibited a distinct preference for binding to several linear trimannoses without the necessity of a branched arm, which differs from the glycan microarray results.

The observed discrepancy could likely be attributed to the nature of the carriers for glycan epitopes in both methods. The GPCs self-assembled into nanoribbons through non-covalent interactions, providing more flexible glycan chains for interaction with glycan-binding proteins when compared to the immobilized glycans on microarray chips. Nevertheless, these results, along with the data for MR and Langerin (Figures 5a and S6c,d) consistent with previous observations,[11c–d,32] indicate that the self-assembling GPC system effectively presents glycan epitopes as clusters necessary for interactions with glycan-binding proteins, and offers valuable complementary information to other methods including the glycan microarray approach. In addition to confirming the binding capacities of known receptor-binding oligomannose epitopes (Figure 5b,c, Table S1, highlighted in blue), our data unveiled a number of oligomannose epitopes that exhibited high affinity towards the receptor proteins (Figure 5b,c, Table S1, highlighted in orange), providing lead compounds for subsequent biological evaluations.

In vitro biological evaluation

High mannose-type *N*-glycans play crucial roles in pathogen endocytosis and immune cell activation by binding to C-type lectin receptors (CLRs) including as DC-SIGN, MR, and Dectin-2.[33] Based on the results of the *in vitro* binding assay, we selected the glycopeptides **F2m1**, **F3m1**, **F3m2**, **F3m4**, and **F3m5**, which exhibited high binding affinity for CLRs, to investigate the immunostimulatory effects of their nanoribbon forms. The immune system is commonly activated through the uptake and presentation of antigens by antigen-presenting cells (APCs).[34] Therefore, we first evaluated the ability of selected samples to promote endocytosis of the ovalbumin (OVA) antigen[35] using a representative monocyte cell line (RAW 264.7). The macrophage cells were incubated for 12 h with either fluorescently labeled model antigen (OVA-Cy3) alone or in combination with peptide **F0g** or GPCs (Figure 6). Flow cytometry analysis revealed that the GPCs significantly enhanced the uptake efficiency of OVA, showing approximately 3–5 fold higher uptake compared to the groups treated with non-glycosylated peptides or OVA alone (Figure 6b). Interestingly, co-delivery of FITC-labeled GPCs (GPCs-FITC) and OVA-Cy3 resulted in approximately 60–90 % of cells positive for both FITC and Cy3 signals (Figures 6a and S13a), which was up to 70 % higher than the group treated with FITC-labeled **F0g** (**F0g-FITC**, Figure S13a,b). This indicated a significantly improved co-delivery capacity induced by the GPCs. Confocal microscopy further con-

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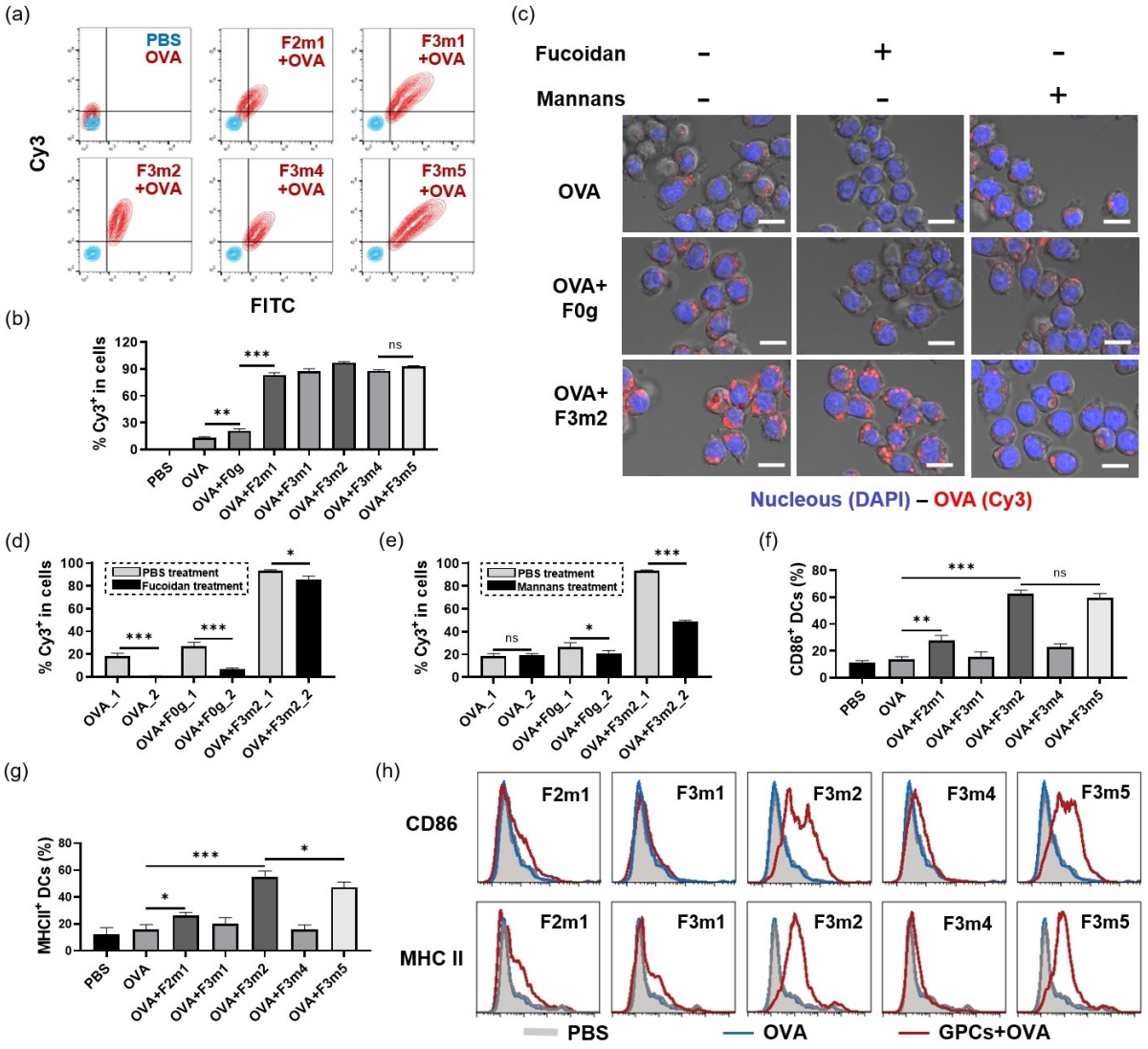


Figure 6. GPC adjuvants enhance antigen endocytosis, processing and presentation in immune cells. a) Flow cytometric plots showing RAW 264.7 macrophages positive for GPCs labeled with FITC and OVA labeled with Cy3. b) Percentage of OVA positive RAW 264.7 macrophages after incubation with OVA-Cy3 alone or in combination with GPCs ($n=3$). c) Fluorescence imaging and d,e) flow cytometric analysis of OVA-Cy3 internalization in RAW 264.7 macrophages ($n=3$) in the absence or presence of different inhibitors, including fucoidan (300 μg/mL) and mannans (300 μg/mL).The percentage of BMDCs positive for f) CD86 and g) MHC-II and h) flow cytometric histogram after incubation with the mixture of GPCs and OVA ($n=3$). Values are the mean ± sd. Statistical significance (one-way ANOVA): *P < 0.05, **P < 0.01, ***P < 0.001. Scale bar: 10 μm.

firmed the co-localization of fluorescent signals within RAW 264.7 cells treated with the mixtures of GPCs-FITC and OVA-Cy3 (Figure S16), as well as in other antigen-presenting cell lines such as bone marrow-derived dendritic cells (BMDCs) and DC2.4 (Figures S17 and S18).

To gain further insight into the mechanism underlying the endocytosis-promoting effect, we examined the uptake of OVA in the presence or absence of GPCs, along with representative receptor inhibitors. Confocal microscopy and flow cytometry analysis indicated that the presence of fucoidan, the scavenger receptor inhibitor, inhibited the cellular uptake of OVA, and the inhibitory effect could be rescued by the addition of **F3m2** (Figure 6c,d). In contrast, the addition of mannans, the CLR inhibitor, significantly reduced the internalization of OVA in the **F3m2**-added group (Figure 6c,e), while it had no significant impact on the groups treated with OVA alone or the mixture of OVA and peptide **F0g**. These results suggest that the endocytosis-promoting effects of the GPC samples may be mediated through the CLRs on immune cells. It is important to note that, at this stage, the involvement of other active endocytosis pathways cannot be completely ruled out, as previous studies have reported that self-assembling peptide hydrogel nanomaterials can also enhance antigen uptake by APCs.[36]

Following endocytosis, APCs play a critical role in the subsequent processing and presentation of antigens, which is essential for activating naive T cells. Therefore, the effectiveness of a adjuvant also relies on its ability to influence the antigen processing and presentation. As APCs mature, they upregulate the expression of the costimulatory

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molecules such as CD86 and major histocompatibility complex (MHC).[34,37] To assess the impact of GPCs on antigen processing and presentation, we measured the expression levels of CD86 and MHC-II on BMDCs treated with OVA, with or without the addition of GPCs. Flow cytometry analysis indicated that the groups containing **F3m2** and **F3m5** displayed significantly higher expression levels of CD86 and MHC-II compared to the OVA-only group, with approximately a 2–3 fold increase (Figure 6f–h). These data together, along with the evidence that no apparent cytotoxicity of the GPCs on tested immune cells (Figure S20a–c), suggest that certain GPCs evaluated in this study have the potential to be safe candidates for further evaluating their immunostimulatory effects *in vivo*.

Immunostimulating effects in mice

Encouraged by the promising *in vitro* results, we further evaluate the efficacy of the superior GPCs, **F2m1**, **F3m1**, **F3m2**, **F3m4**, and **F3m5**, as vaccine adjuvants in mice. Interestingly, during sample preparation, we observed that mixing the glycopeptides with the OVA antigen resulted in a stickier suspension, displaying thicker and more tangled nanofibers under TEM (Figure S21). This may be attributed to potential interactions between the GPC nanostructures and OVA proteins. However, upon closer inspection of the individual bundles of formed nanofibers, the morphology remains akin to the nanoribbons observed in our prior experiments. To investigate whether this morphological change would impact the sample preservation time in vivo, we subcutaneously injected the mixture of **F3m2** and Cy3-OVA and monitored it using an *in vivo* imaging system (IVIS). The imaging results indicated that the addition of **F3m2** extended the in vivo retention time of OVA from 2 h to approximately 2 days (Figure S22), comparable to the aluminum adjuvant.[38]

Furthermore, we evaluated the immunostimulatory capabilities of the GPCs **F2m1**, **F3m1**, **F3m2**, **F3m4** and **F3m5**, which were co-incubated with OVA in water overnight at 4 °C and subsequently injected into C57BL/6J mice. For comparison, three groups of mice were subcutaneously given PBS, OVA alone, and OVA absorbed on aluminum. Injections were administered on days 0, 7, 14, and 21, and mouse sera were collected on day 28 to measure antibody titers (Figure 7a). As shown in Figure 7b, the groups containing GPCs elicited significantly enhanced anti-OVA IgG titers compared to the group injected with OVA alone. Among all the tested GPCs, **F3m2**, which appends α-1,2-α-1,3-mannotriose, and **F3m5**, which appends α-1,6-α-1,6-mannotriose, exhibited the most pronounced immunostimulating effects, resulting in an anti-OVA IgG titer approximately 6-fold higher than that of the group with OVA alone (Figure 7b). This result also aligns with the in vitro BMDCs activation experiments. Importantly, all glycopeptides showed no production of IgG titer against themselves (Figure S24), and no significant cytotoxicity were observed to the spleen cells (Figure S20d).

To gain further insight into the immune-activating mechanisms, we analyzed the IgG subclasses using ELISA. Compared to the group treated with OVA alone, the groups supplemented with GPCs generally elicited higher levels of anti-OVA IgG1 (Figure 7c). Noticeably, only the groups containing **F3m2** or **F3m5** showed an additional capability to enhance IgG2a and IgG2b production (Figure 7d,e). This finding suggests that **F3m2** and **F3m5** may be capable of inducing Th1-biased immune responses in addition to the Th2-biased pathway, which aligns with their ability to promote DC maturation (Figure 6h). To further validate this conclusion, we measured the cytokines associated with the Th1 and Th2 pathways.[36a] The GPC-treated groups exhibited higher levels of Th2-relevant IL-5 production compared to the OVA-only group (Figure 7f), which subsequently promoted the IgG1 production by B cells. As a comparison, only the groups with **F3m2** and **F3m5** demonstrated a significantly stronger ability to stimulate IFN-γ secretion than all other groups (Figure 7g), consistent with their effects on Th1-related IgG2a and IgG2b production.

Additionally, considering the functional heterogeneity of T cell immunity, where $CD4^+$ and $CD8^+$ T cell subsets contribute to both protective immunity and long-term immunological memory, we investigated the differentiation status of $CD4^+$ and $CD8^+$ T cells induced by the OVA vaccination with GPC adjuvants. The results revealed a significant increase in $CD4^+$ T cells in mice immunized with mixtures of OVA and GPCs (**F2m1**, **F3m1**, **F3m2**, **F3m4** or **F3m5**) compared to the OVA-only group (Figure 7h–i). Remarkably, among all the groups, those supplemented with either **F3m2** or **F3m5** exhibited higher levels of $CD8^+$ T cells (Figure S25), presumably associated with the additional induction of Th1 cytokines. The ability of these two GPCs to stimulate $CD8^+$ T cell proliferation would be particularly advantegeous for effective vaccine adjuvants for tumor treatment since $CD8^+$ T cells are potent effectors in anticancer immune responses.[39] Taken together, our data demonstrate the excellent potential of the GPCs as adjuvants in vaccine development.

Conclusion

In summary, we have developed a self-assembling system based on glycopeptide conjugates, enabling the generation of uniform nanostructures, such as nanoribbons. This system offers a versatile platform for investigating the structure–activity relationships of various O-linked oligosaccharides through direct application in cell-based assays and animal models. This feature represents a potential advantage over traditional glycan arrays, and opens up possibilities of applications in translational researches, such as therapeutic developments. The utility of this self-assembling system has been further demonstrated through the identification of mannosylated GPCs **F3m2** and **F3m5**, which effectively promote the maturation of DCs and enhance antigen presentation, leading to potent immunological responses in mice. Importantly, the **F3m2** and **F3m5**-derived samples not only elicit IgG titers comparable to those induced by

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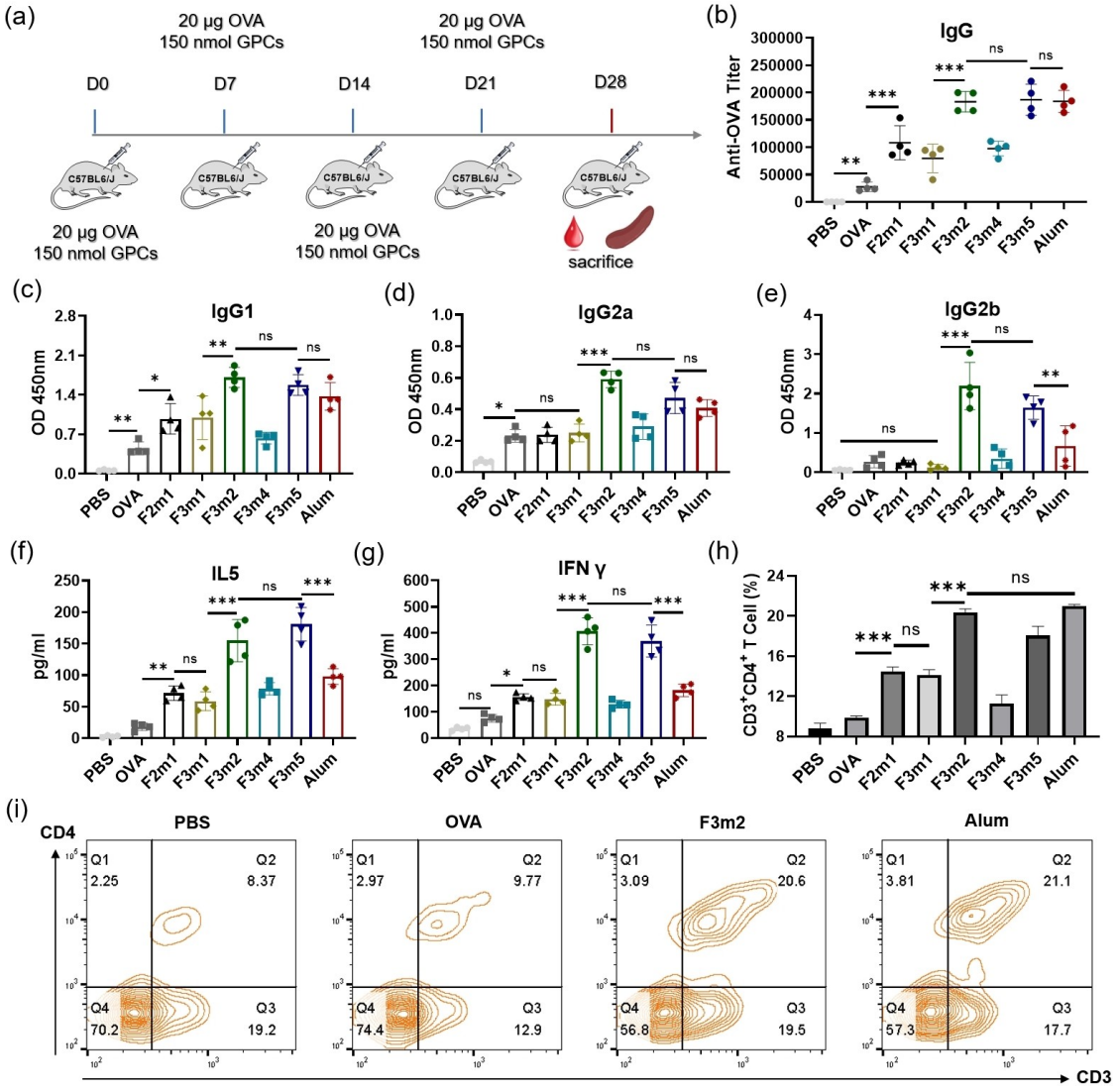


Figure 7. Effect of GPC-based vaccination on antigen-specific antibody production and humoral and cellular immune responses. a) Immunization schedule for GPC-based vaccination in C57BL6/J mice. C57BL/6J mice were subcutaneously immunized with different vaccines, with OVA, Alum, and GPCs administered at does of 20 μg, 200 μg, and 150 nmol per mouse, respectively. b-e) Detection of anti-OVA IgG, IgG1, IgG2a, and IgG2b antibodies in plasma on day 28 using enzyme-linked immunosorbent assay (ELISA) ($n=4$). Measurement of f) IL-5 and g) IFN-γ production in the supernatants of splenocytes from immunized mice using ELISA ($n=4$). h) Percentages of CD3 and CD4 positive splenocytes and i) the corresponding flow cytometric plots ($n=3$). Values are the mean ± sd. Statistical significance (one-way ANOVA): *$P<0.05$, **$P<0.01$, ***$P<0.001$.

aluminum adjuvants with OVA antigen but also exhibit significantly higher expression of IFN-γ and IL-5 compared to aluminum adjuvants, suggesting a stronger ability in activating innate immunity. Therefore, beyond elucidating the preference of α-1,2-α-1,3- and α-1,6-α-1,6-linked mannotrioses within high mannose-type N-glycans in interacting with DC-SIGN and highlighting structural significance of oligosaccharides as ligands, our study also provides lead compounds comprising biologically compatible components with structural modularity. These glycopeptide conjugates can be further optimized to discover safe and effective vaccine adjuvants with enhanced immunostimulatory properties.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: Glycopeptide Conjugates · Immunostimulatory Activity · Nanoribbons · Self-Assembly · Vaccine Adjuvants

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